



GROW_SCI_A2_W07 · GROW Science teacher note

Staff-facing; review only. Verbatim SoW objective: Review and evidence autumn science (Entry Level Science / UAS).

Source: GROW scheme of work, Autumn 2 week 7. Pupil goal: use a model and a matching clue to show what you know in science.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40.

One-minute resets are inside We do 2, Review and Exit. Success: I can choose a model that fits my question., I can give a clue that supports my answer., I can name one next step after feedback.

Retrieval and I-do check

When	Question	Answer	SoW cell
last lesson	What changed in last week's fair test?	The surface; the toy, start and launch were controls.	GROW scheme of work, Autumn 2 week 6
2-4 weeks	What model can help with a toy motion question?	Toy and surface.	GROW scheme of work, Autumn 2 week 5
last term	Name one force you learned in Autumn 1.	Gravity, air resistance or friction.	GROW scheme of work, Autumn 1 week 1

Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: Show me the model you would use for day and night. Answer: Torch and turning ball. All pupils show by point, card or board.

Words (same as page and organiser):

Word	Tier	Simple definition	Example sentence
evidence	Tier 2	a result or observation that supports a claim	A measured distance is evidence about a toy run.
force	Tier 3	a push or pull on an object	Friction is a contact force.
rotation	Tier 3	a turn around an axis	Earth's rotation makes day and night.
climate	Tier 3	weather patterns over many years	Climate is not the weather on one day.

Faded example, tiered task and stretch

I do full: "Question: why does a place get night? → choose torch and turning ball → marked place turns away from light → explain Earth's rotation." We do part-done: "Question: what slowed the toy? → choose toy and surface → observe → explain a possible force; check with repeat runs." Removed step: Identify an observed travel-distance clue for the toy claim, not a space fact. Independent: no filled step. Next fade: Next time remove the supplied model choice in practice and ask the pupil to select it from three familiar kits.

We do F: Point to the torch and the marked place before naming the clue. M: Use a model and matching clue to explain a science idea. H: Explain why a correct model can still leave out size, distance or other details. Independent F: Match one question to its model. M: Use a model and



matching clue to explain a science idea. H: State one limit of a model and suggest a further check. Go deeper: For one claim, give two checks and one model limit. Say which extra observation you would seek. Named harder stretch — Two ways to check: Compare two model limits. Choose a claim neither can settle alone and name a new observation that would help.

The model, source or practical route is: toy and surfaces, torch and ball, long-term climate cards. Two on-screen interactions are card sort and Which clue fits a day-and-night claim?; native keyboard controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Hinge and decision

Question: A new question asks why a model sled stops sooner on a mat. Which kit can help test it?

Option	Correct?	Diagnosis
Sled and two marked surfaces.	yes	Uses the new case accurately.
A torch and turning globe.	no	Uses a rotation model to answer a friction question.
Two ice cubes on dark card.	no	Uses melting ice rather than a moving sled to test the surface effect.
A weather report from today.	no	Uses a single weather report instead of a surface-and-motion test.

Rule: About 3 in 4 or more right → move on. Otherwise re-model with the named I-do example, then ask the parallel hinge. Parallel: A new place on a globe turns into shade. Which kit explains that: globe and fixed lamp, or sled and mat? Answer: globe and fixed lamp.

Misconceptions:

Misconception	Spot	Correct
A melting-ice model tests why a moving sled stops on a surface.	Pupil chooses C, an ice model for sled motion.	Use the sled with two marked surfaces for the process in the question.
One day's weather report can explain the sled's stopping distance.	Pupil chooses D, using one day's weather to explain the sled.	Observe and compare the sled on surfaces instead of using unrelated weather.

AQA Entry Level Science style bridge (classroom planning): Use a suitable model to explain a science idea with a matching clue. This is a planning bridge, not a claimed unit outcome or code.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ?. At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ?. R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations

Broad area	EEF five-a-day strategy	Specific support
Cognition & Learning	scaffolding	Offer three familiar model cards with one question highlighted, then fade the highlight.



Broad area	EEF five-a-day strategy	Specific support
Communication & Interaction	cognitive/metacognitive strategies	Allow verbal, signed or AAC practice of claim and clue before the independent response.
SEMH	flexible grouping	Agree a low-voice check-in and known pause; do not create pressure to perform publicly.
Sensory/Physical	technology	Keep model kit fixed and high contrast; offer TTS and agreed accessible evidence mode.
EAL (language, not SEN)	explicit instruction	Pre-teach the same four word icons. Keep M/H challenge open with language frames and planning in any language.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Wait and count to 5. → Point to the lesson's diagram, kit or source. → Ask: What is done, and what is the next step? → Model one step, then return the next step to the pupil. → Finish one example together, then fade the help. No physical prompt. Stage scripts: I do "Watch me select one model for this question."; We do "What is supplied? Which observed clue is missing?"; Independent "Choose your own question. Which clue can you show without my prompt?"

At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Cold-call / invite bank, answers:

- F: Which model fits day and night? → Torch and ball.; Which model fits toy motion? → Toy and surface.
- M: What makes a clue useful? → It matches the question and claim.; How can you show a force result? → Point to or name an observed toy run, with controls noted.
- H: What is one limit of a torch-and-ball model? → It is not to scale, among other limits.; What would strengthen a climate claim? → Long-term observations, not one weather report.

Regulation: Agree a quiet pause signal and supervised place. Check in before hinge and Independent; point to the last completed step on re-entry. No evidence while a pupil is distressed.

Lesson record: GROW_SCI_A2_W07 via staff device or projector only. Gatsby 4: Laboratory technician: organise a result so another person can check a claim.

Reading gate: GROW 8-11; rough visible-body ARI grade 4.63; age guide 10.63 years; rough FK grade 4.62; 821 pupil-visible words excluding marked adult guidance and navigation. Screening estimate only. This is a rough screen, not an individual assessment.

Prepared model handling · practical safety

Kit: Adult-prepared familiar model kit for the selected question

Method: Use only the prepared model selected for the agreed practice or formal task.

Pupil Risk: Loose model parts, a toy launch or a torch beam could cause harm if misused.



Pupil Safe: Keep model parts on the tray, launch only with adult permission and aim a torch away from eyes.

Staff confirm the local school risk assessment and exact kit; this does not assert a SoW core practical.

SLT planning and evaluation

Lesson code: SCI-GROW-A2-W07

SoW reference: GROW scheme of work, Autumn 2 week 7

Exact SoW outcome: Review and evidence autumn science (Entry Level Science / UAS).

Evidence (scheme of work): Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-GROW-A2-W07. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Choose a model used this term. Point to one question it could help answer. Staff: Offer a predictable seat or standing spot and the first visual cue. No public test.

Starter (3 min): Think back. Say, point or write one answer for each question. Staff: Ask three low-stakes retrieval questions; prior coverage must be checked.

I Do (5 min): I choose a question, then a matching model and clue. For day and night, the torch lights one side of a turning ball. I explain the place turning into shade. This is practice, not help during formal evidence. Staff: Model the full reasoning step, pre-teach four words and ask the show-me question.

We Do (7 min): In practice, the question and torch-and-ball model are supplied. Complete the missing clue: what happens to the marked place as the ball turns? Staff: Show the part-done example, then allow paired talk before an answer. One-minute reset inside this stage.

We Do 2 (5 min): Show everyone which model best helps explain day and night. Staff: All pupils answer the four-option hinge; use the decision rule, not a score.

Independent (8 min): Choose two science questions from forces, space or climate. Show an answer with a model and a matching clue. Use only your agreed support if this is formal evidence. Staff: One adult supports agreed pupils, the other holds the group; swap across the week. One-minute reset inside this stage.

Review (4 min): In practice, hear one comment on claim and clue. Self-edit one mismatch. If this is formal evidence, receive and act on feedback after collection under the agreed conditions. Staff: Give a specific verbal comment (VF) and invite one self-edit (/) or read-back (?).

Exit (3 min): Tell the adult what you learned, which clue you used, your feedback and one next step. They use your answer to plan the next model or support. Staff: Hear the answer and name how it changes the next example or support. One-minute reset and handoff.

Planned adaptation — cognition learning: Offer three familiar model cards with one question highlighted, then fade the highlight.

Planned adaptation — communication interaction: Allow verbal, signed or AAC practice of claim and clue before the independent response.



Planned adaptation — semh: Agree a low-voice check-in and known pause; do not create pressure to perform publicly.

Planned adaptation — sensory physical: Keep model kit fixed and high contrast; offer TTS and agreed accessible evidence mode.

Planned adaptation — eal: Pre-teach the same four word icons. Keep M/H challenge open with language frames and planning in any language.

Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: cognitive/metacognitive strategies; semh: flexible grouping; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Choose two science questions from forces, space or climate. Show an answer with a model and a matching clue. Use only your agreed support if this is formal evidence. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Spring 1 W01. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Spring 1 W01: Compare and group materials by properties (hardness, conductivity, solubility).

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Identify an observed travel-distance clue for the toy claim, not a space fact. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A new question asks why a model sled stops sooner on a mat. Which kit can help test it?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Choose a model used this term. Point to one question it could help answer. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Choose two science questions from forces, space or climate. Show an answer with a model and a matching clue. Use only your agreed support if this is formal evidence. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use evidence, force, rotation, climate with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.



Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.

Working Scientifically sources: AQA WS 1.2 (models) and WS 1.3 (their limitations),
<https://www.aqa.org.uk/subjects/science/gcse/science-8464/specification/working-scientifically>.